CS201 Data Structures - Course Syllabus

Credits: 4 (3 lecture + 1 lab). Prerequisite: CS101 Programming Fundamentals.

Unit 1: Introduction and Arrays

Abstract data types, asymptotic notation (Big-O, Omega, Theta), arrays, dynamic arrays and amortised analysis.

Unit 2: Linked Lists, Stacks and Queues

Singly, doubly and circular linked lists; stack and queue ADTs; applications such as expression evaluation and BFS.

Unit 3: Trees

Binary trees, tree traversals (inorder, preorder, postorder), binary search trees, AVL trees and rotations, heaps and priority queues.

Unit 4: Graphs

Graph representations, depth-first and breadth-first search, shortest paths (Dijkstra), minimum spanning trees (Prim and Kruskal).

Unit 5: Hashing and Sorting

Hash tables, collision resolution, quicksort, mergesort, heapsort and lower bounds for comparison sorting.

Textbook: Cormen et al., Introduction to Algorithms, 4th edition.

CS201 Lab Manual - Experiments

Lab experiments are evaluated for 25 marks: 15 marks for weekly records and 10 marks for the final lab examination. Students must complete at least 8 of the 10 experiments to be eligible for the lab examination.
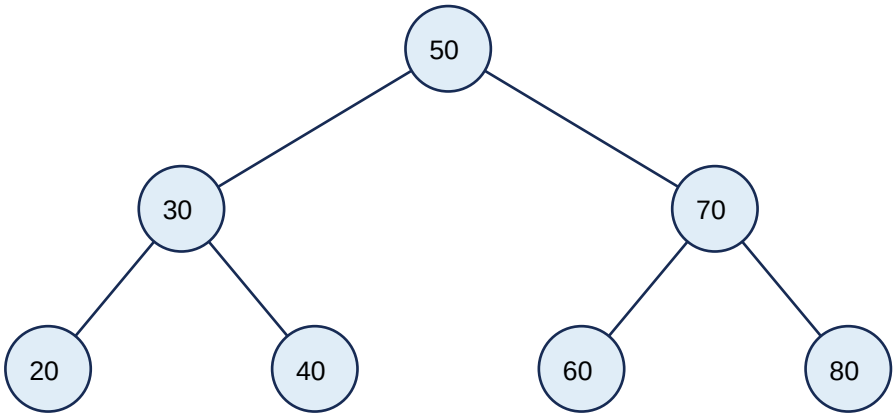
Figure 2: Binary search tree used in Experiment 6 (insertion and deletion).